

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

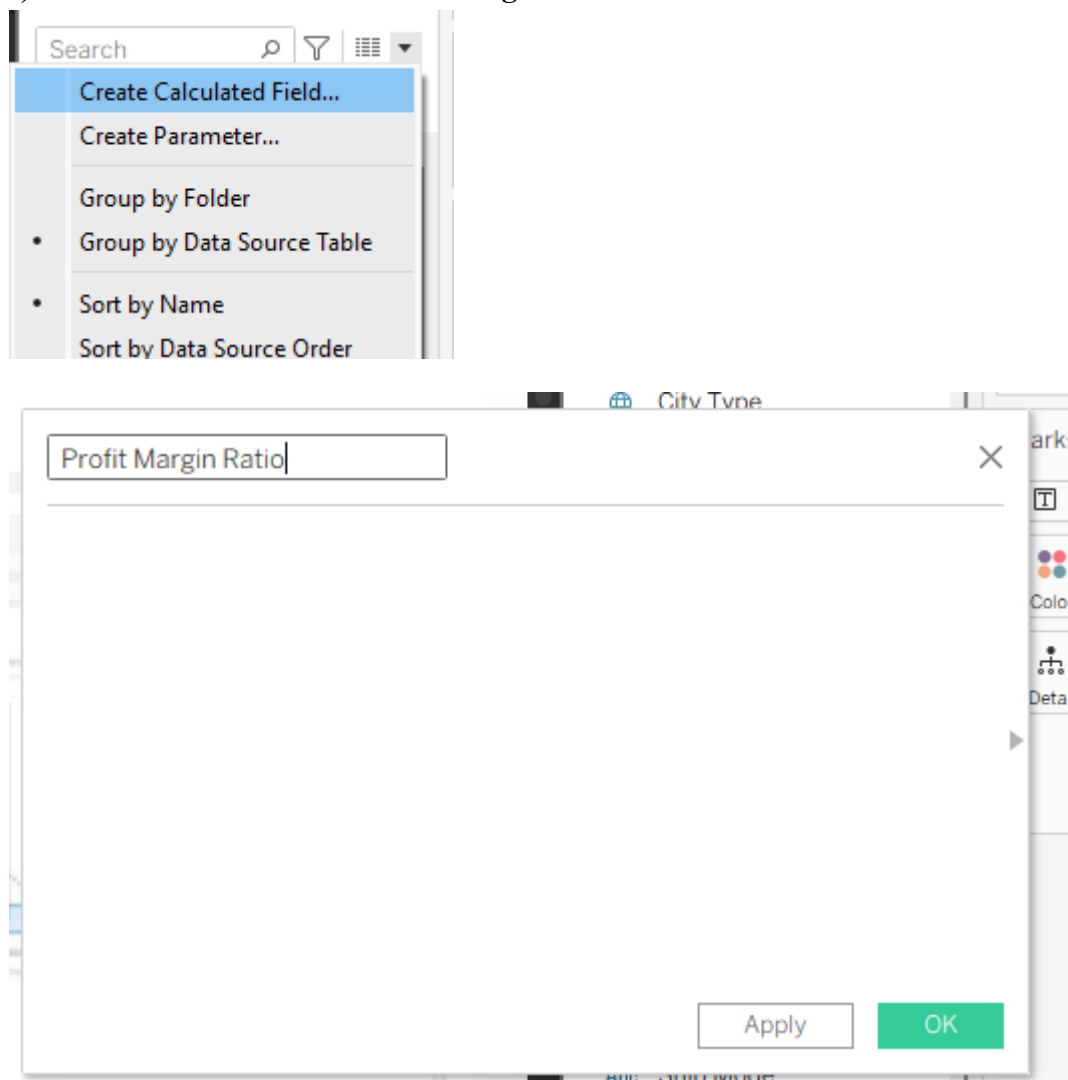
Practical – IV

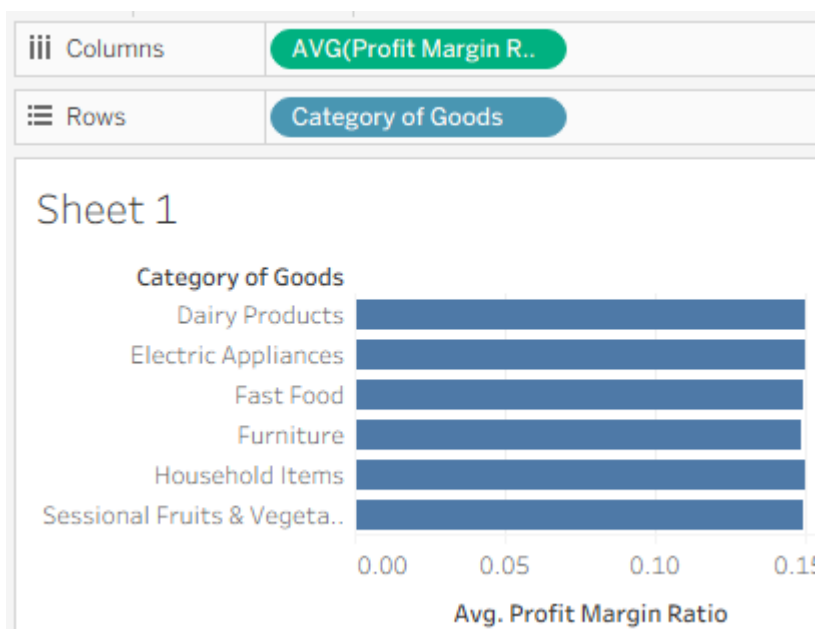
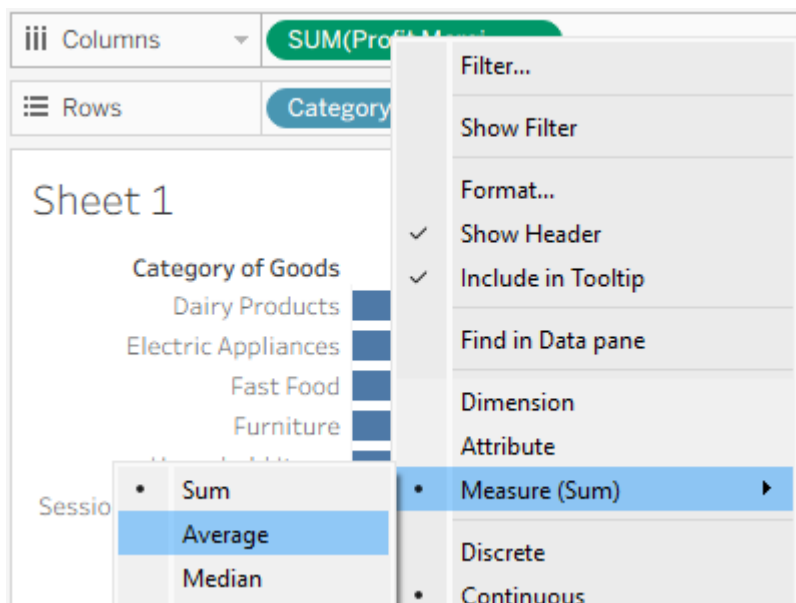
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Class: T.Y.IT	Batch: 1
Date of Assignment: 20-08-2026	Date/Time of Submission: 20-08-2026

Part B – Program

Performing Data Analysis using Calculations in Tableau

a) Create basic calculated fields using formulas





b) Perform row-level calculations

Total Revenue Before Discount

×

[Sales] * [Quantity]

The calculation is valid.

Apply

OK

Columns	SUM(Total Reven..	SUM(Sales)
Rows	Product Name	
Sheet 1		
Product Na..		
Apples - 100		
Apples - 102		
Apples - 103		
Apples - 104		
Apples - 105		
Apples - 106		
Apples - 107		
Apples - 108		
Apples - 109		
Apples - 110		
Apples - 111		
Apples - 112		
Apples - 113		

c) Perform aggregate-level calculations

Aggregate Profit Margin

×

$$\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$$

The calculation is valid.

Apply

OK

Columns	AGG(Aggregate Profi..
Rows	Sub-Category

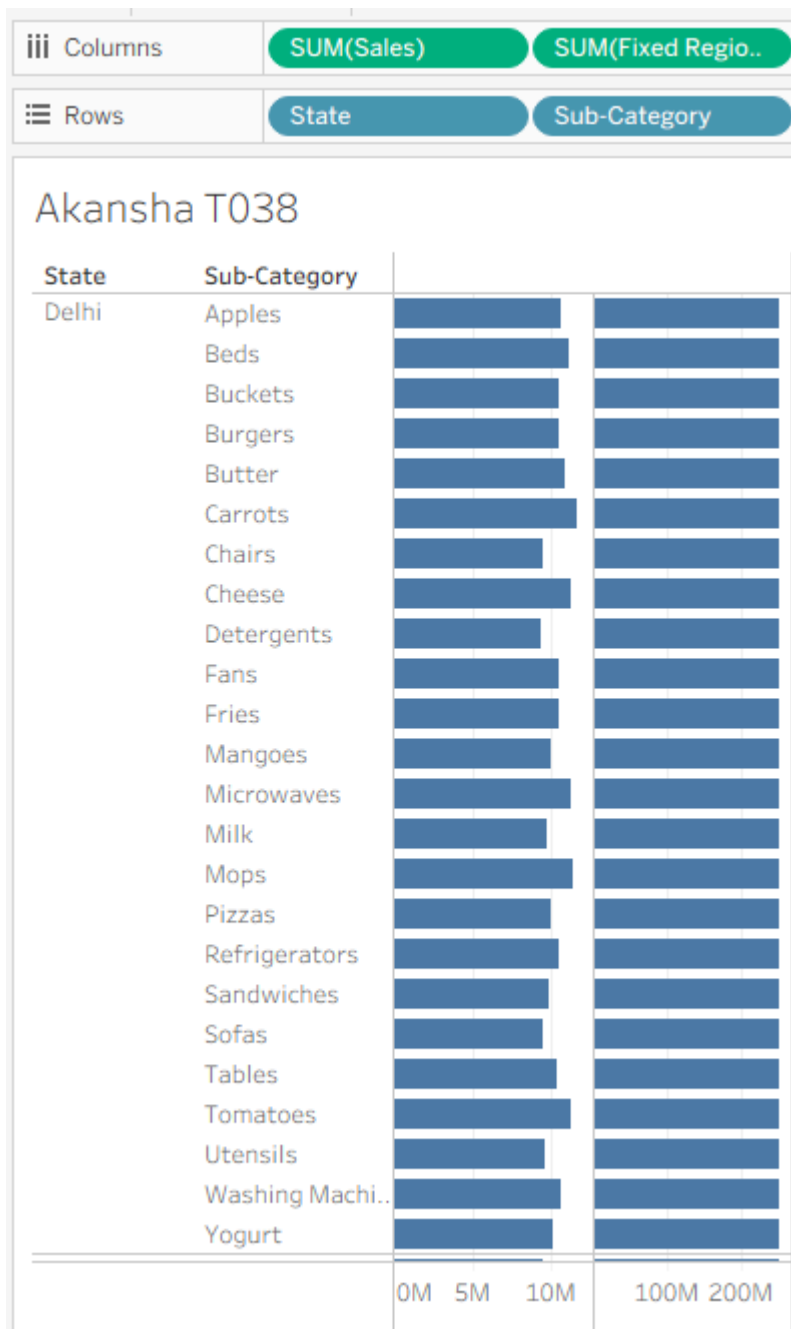
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Sub-Category	
Apples	
Beds	
Buckets	
Burgers	
Butter	
Carrots	
Chairs	
Cheese	
Detergents	

d) Implement FIXED Level of Detail (LOD) calculations

Fixed Regional Sales

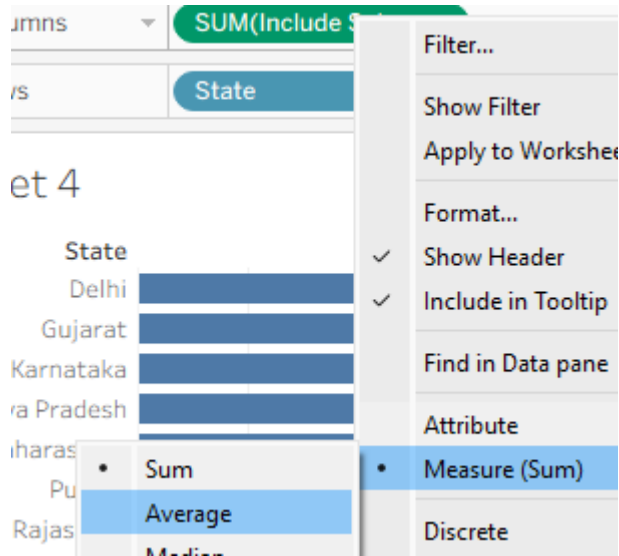
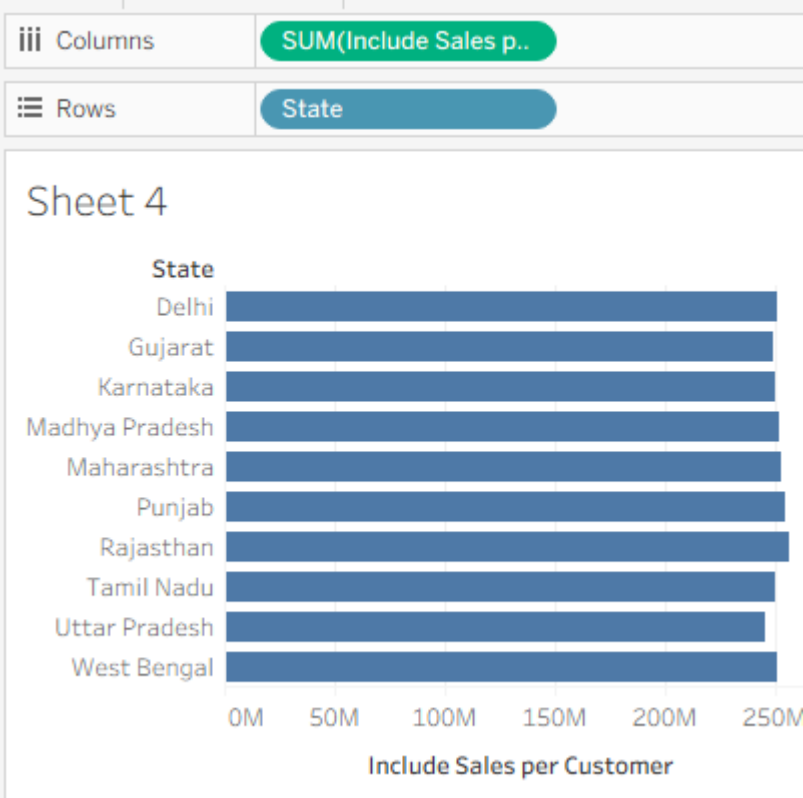
$$\{ \text{FIXED } [\text{State}] : \text{SUM}([\text{Sales}]) \}$$



e) Use INCLUDE and EXCLUDE LOD expressions

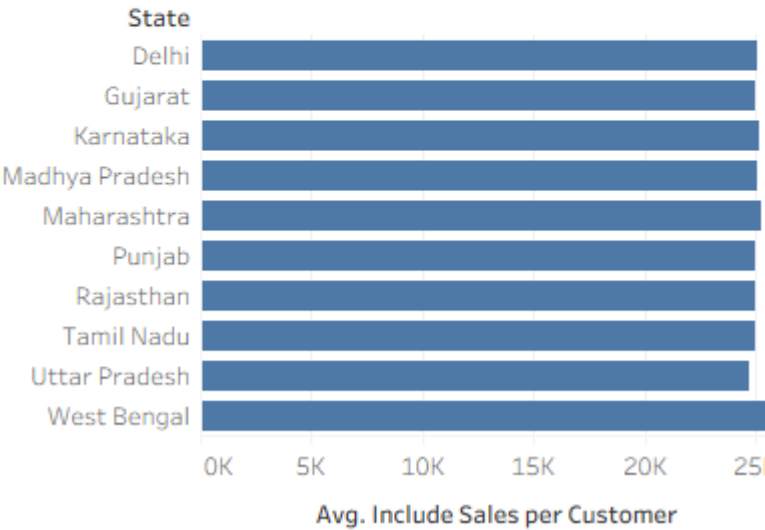
Include Sales per Customer

```
{ INCLUDE [Customer ID] : SUM([Sales]) }
```



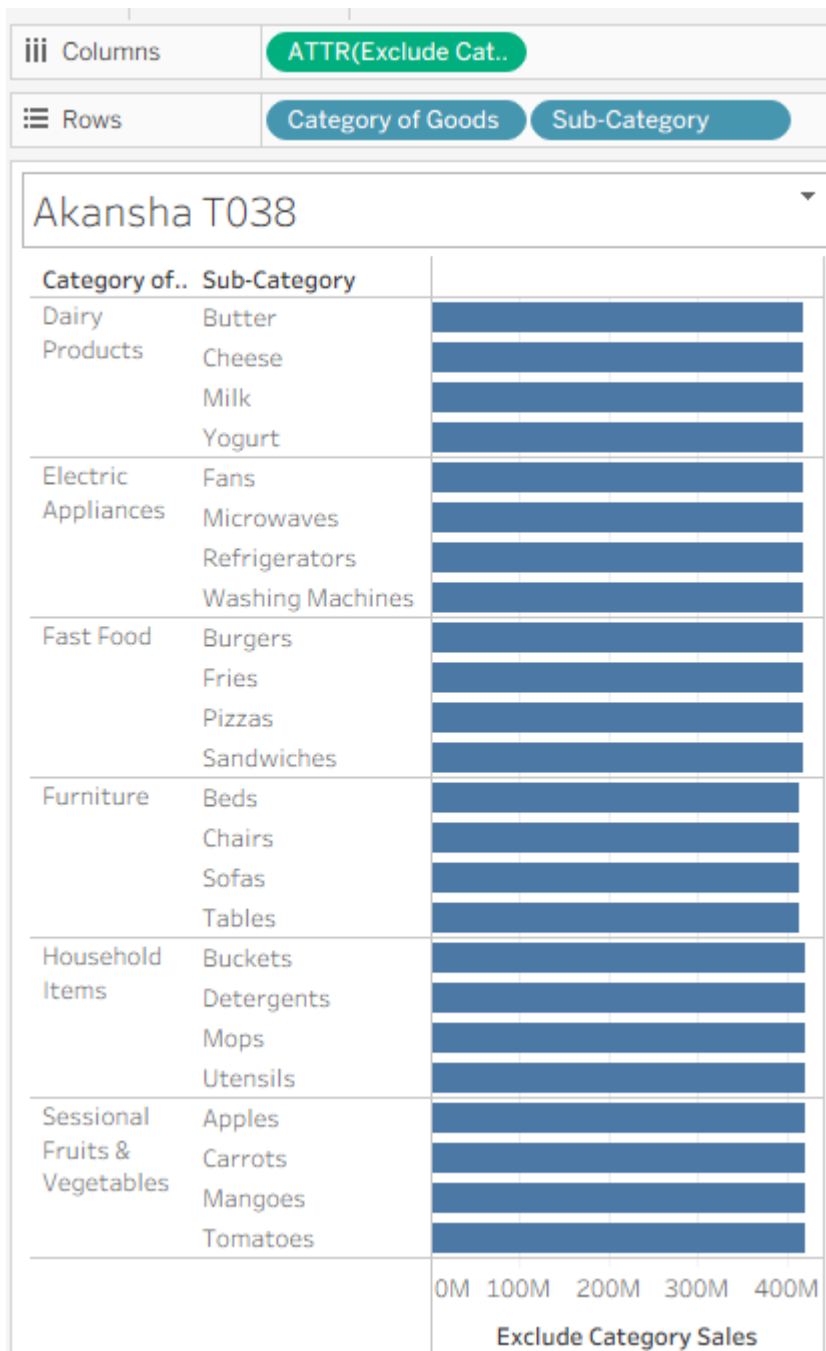
Columns	AVG(Include Sales p..
Rows	State

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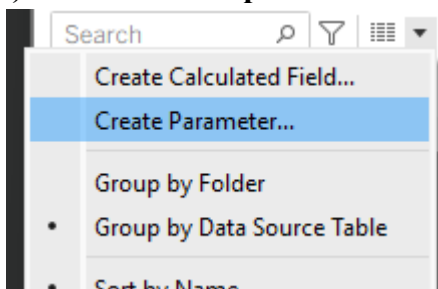


Exclude Category Sales

{ EXCLUDE [Sub-Category] : SUM([Sales]) }



f) Create and use parameters for dynamic analysis



data (2)

Create Parameter

×

Name

Select Profit Threshold

Properties

Data type

Float

Display format

500

Current value

500

Value when workbook opens

Current value

Allowable values

☒ All

☐ List

☐ Range

Cancel

OK

```
# store sales data (2).C...
```

Parameters

Select Profit Threshold

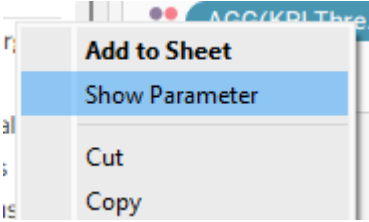
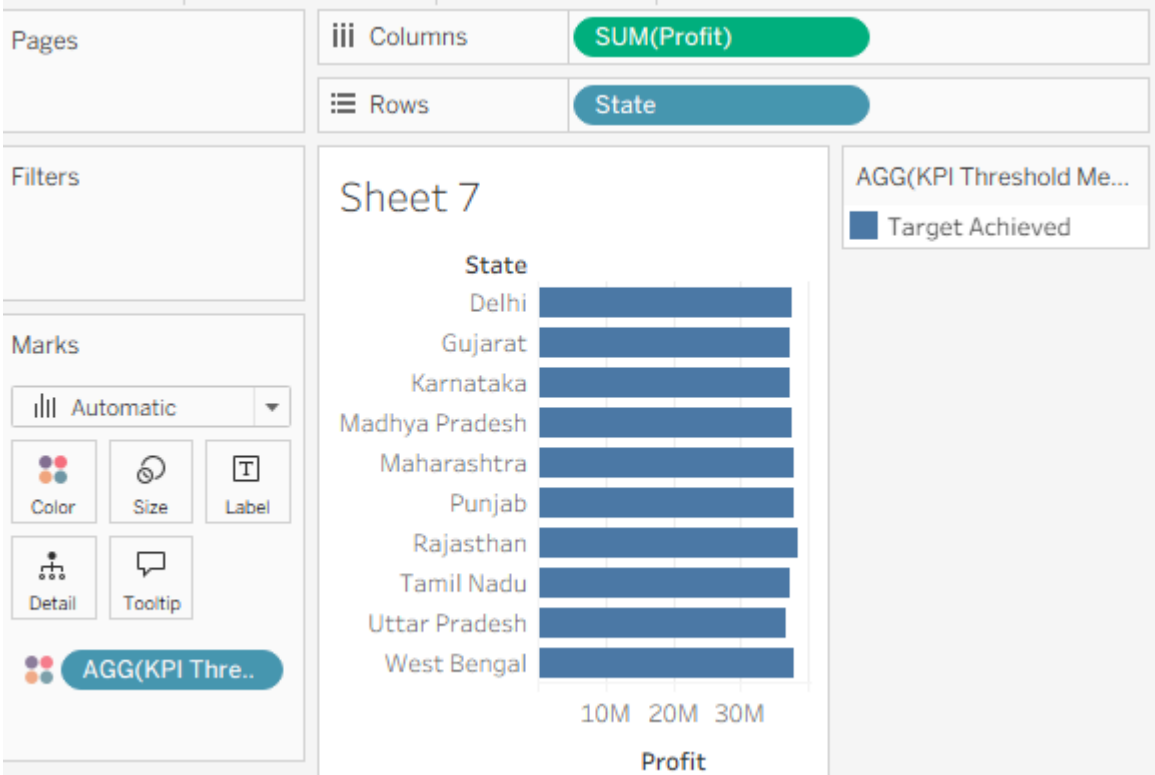
KPI Threshold Met?

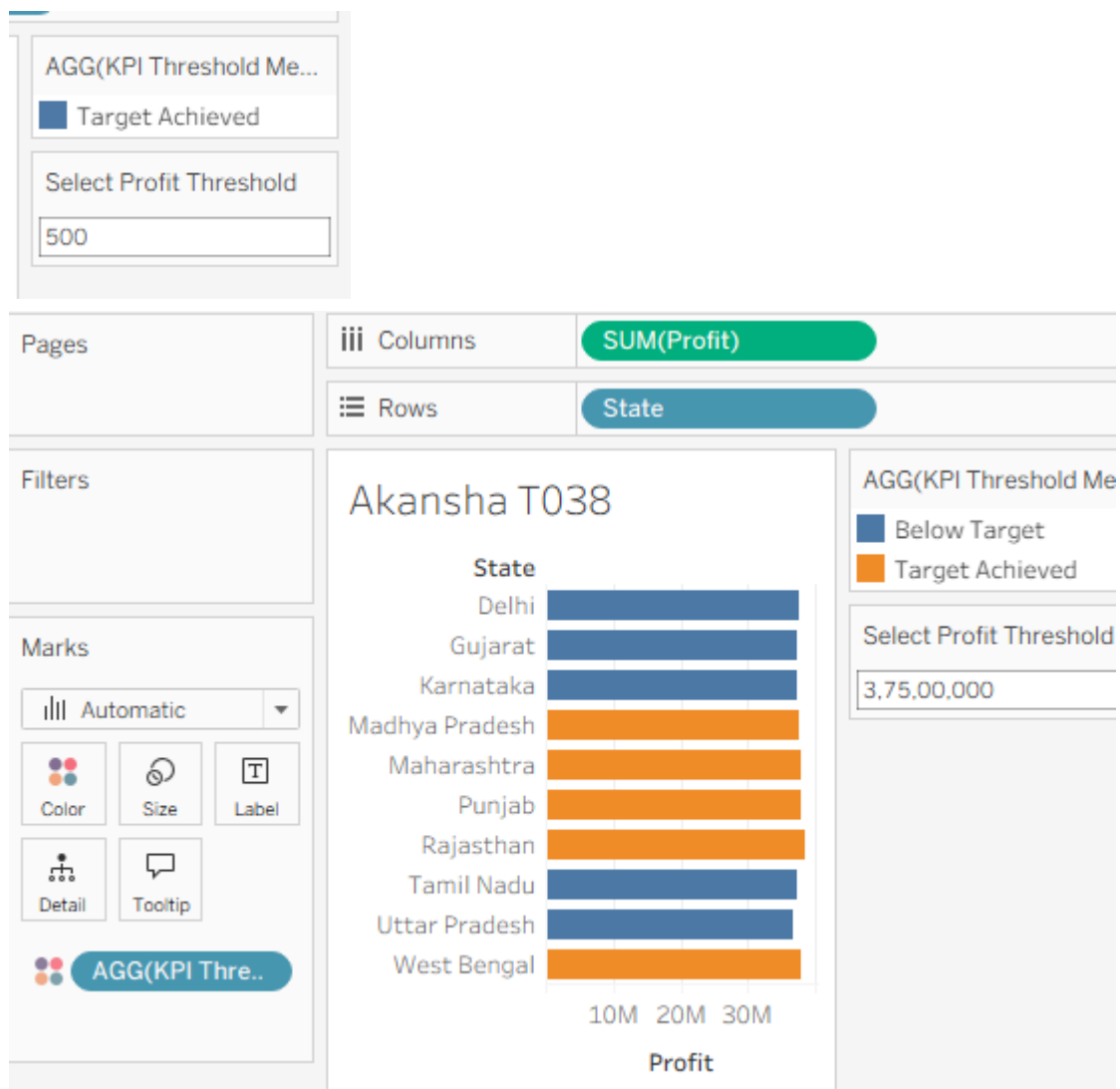
×

IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Ac
ELSE "Below Target"
END

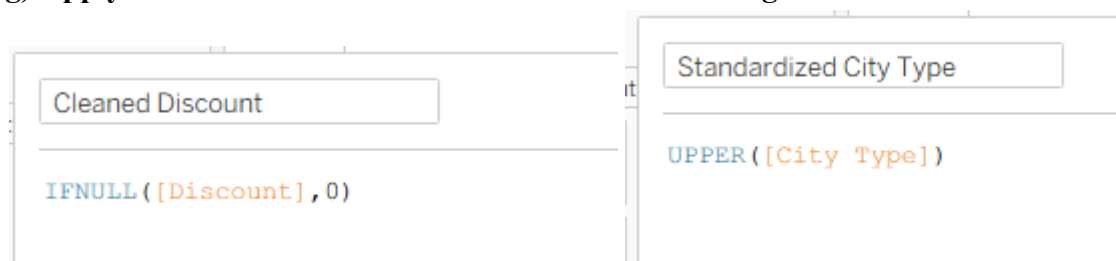
×

END "Target Achieved"





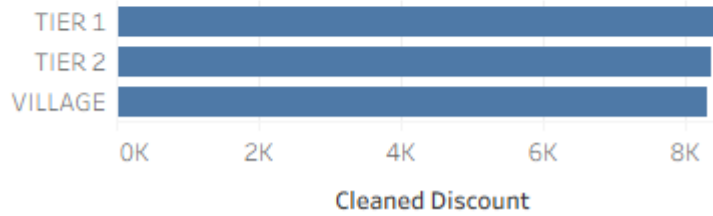
g) Apply calculations to fix data issues and enhance insights



Columns	SUM(Cleaned Discou..
Rows	Standardized City Ty..

Sheet 8

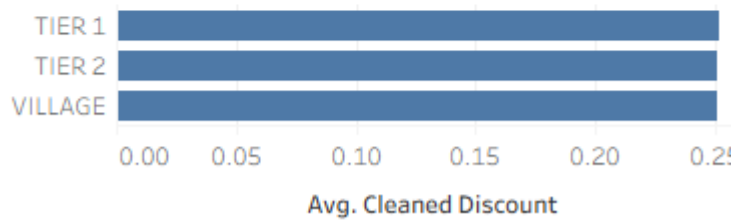
Standardize..



Columns	AVG(Cleaned Discou..
Rows	Standardized City Ty..

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Standardize..



Part C – Curiosity Question

1. Can Tableau calculate a value that is not directly available in the dataset?
2. What happens if we change a calculated field after using it in multiple charts?
3. Why might two calculations using the same Sales field produce different results?
4. Can Tableau calculate the total sales of each customer even when Customer Name is not shown in the chart?
5. What is the difference between calculating Profit for every row and calculating total Profit for a category?
6. Can a parameter allow a user to switch between Sales, Profit, and Quantity in the same chart?
7. What happens to a calculated field when the original field contains NULL values?
8. Can calculations be used to create new categories such as High, Medium, and Low Sales?
9. Why would we use LOD calculations when normal aggregations are already available?
10. Can Tableau calculations help identify the most profitable customer even when the dataset does not contain a “Best Customer” field?
11. What would happen if a FIXED LOD calculation and a normal aggregation were used together?
12. Can calculations be used to correct data problems without modifying the original CSV file?
13. How could you design one chart that allows a user to switch between Sales, Profit, and Order Quantity?
14. If two customers have the same Sales, how could Tableau help determine which one is more profitable?

15. Can calculated fields and parameters make a Tableau dashboard behave like an interactive analytical tool?